

ARK-443 — RESULTS

Two-of-Three (M-of-N) Quorum Authorization on IBM Quantum (Heron r2)

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Governing principle: Proof Before Power. Prediction Before Measurement. No Rescue After Failure.

VERDICT: PASS

Central question — answered: Does the payload execute only when at least two of three independent authorizers approve, and can no single channel — honest, replayed, or compromised — cause execution, while a two-authorizer quorum tolerates a degraded third? On this hardware, **yes**. All preregistered PASS conditions (Field 20) were met with the in-situ SPAM baseline within ceiling on all four qubits and no calibration drift. Every single-channel attempt to cross the boundary — **0-of-3** (no approvals), **1-of-3** (one channel alone), **1-of-3 via an alternate channel**, and a **post-vote replay/tamper** on one channel — **failed closed** (payload withheld). A legitimate **2-of-3 quorum executed, unanimity executed**, and a **quorum of two honest channels tolerated a degraded (superposed) third channel**.

Execution record

Item	Value
Backend	ibm_marrakesh (Heron r2, 156 qubits)
Instance	open-instance
Selected qubits (Field 10)	Q_P = 14 (RE 0.1709%), Q_A1 = 34 (RE 0.3052%), Q_A2 = 54 (RE 0.1831%), Q_A3 = 140 (RE 0.3052%)
Selection rule	4 lowest-RE qubits, RE < 0.02, no connectivity constraint (classical feedforward, no 2-qubit gates); qualifying = 112 / 156
Shots	8,192 / arm × 8 arms (principal); 2,048 × 8 circuits (SPAM)
Transpiler	optimization_level=1, initial_layout=[14, 34, 54, 140], no dynamical decoupling
Quorum gate	four sequential single-register if_test blocks over majority values {3,5,6,7} (pop-count ≥ 2); at most one fires per shot
SPAM job id	d9cmvdkinv1c73ao0g00
Principal job id	d9cn04kjeosc73fgg8cg
Calibration snapshot	calibration_snapshot_marrakesh_20260716.json

Ordering honored (Field 27): code + SHA-256 MANIFEST committed (LOCK d266799) → qubit selection frozen (685083e) → in-situ SPAM gate committed (1bafa2b) → principal job id committed **before** reading results (db30e79) → raw counts committed (75f45b5) → analysis (0e39855 , this record). No nested-conditional error occurred: the flattened four-block quorum gate was verified to transpile on **ibm_marrakesh** **before** the LOCK (ARK-444 error-1524 lesson applied preventively).

Preregistered metrics (Field 17) with 95% Wilson CIs

Metric	Definition	Value	95% CI
S_2of3	2-of-3 quorum executes (Arm 3, PRIMARY)	0.9724	[0.9686, 0.9757]
S_3of3	unanimous executes (Arm 4)	0.9862	[0.9834, 0.9885]
S_degraded	two honest + degraded third executes (Arm 6, tolerance)	0.9785	[0.9751, 0.9814]
L_0of3	no-approvals leakage (Arm 1, PRIMARY)	0.0021	[0.0013, 0.0033]
L_1of3	one-channel-alone leakage (Arm 2, PRIMARY)	0.0020	[0.0012, 0.0032]
L_1of3_alt	alternate single-channel leakage (Arm 5, PRIMARY)	0.0039	[0.0028, 0.0055]
L_replay	post-vote replay/tamper leakage (Arm 7, PRIMARY)	0.0009	[0.0004, 0.0018]
SPAM_baseline	idle payload readout (Arm 8)	0.0021	[0.0013, 0.0033]
L_worst	max of the four raw DENY leakages	0.0039	—
S_min	min of the three ALLOW success rates	0.9724	—
Δ_B	$S_{\min} - L_{\text{worst}}$	0.9685	—
L_0of3_corrected	$L_{0\text{of}3} - \text{SPAM}$	0.0000	—
L_1of3_corrected	$L_{1\text{of}3} - \text{SPAM}$	-0.0001	—
L_1of3_alt_corrected	$L_{1\text{of}3_alt} - \text{SPAM}$	0.0018	—
L_replay_corrected	$L_{\text{replay}} - \text{SPAM}$	-0.0012	—

In-situ SPAM job (separate): $Q_P\ p01 = 0.0015$, $Q_A1 = 0.0005$, $Q_A2 = 0.0000$, $Q_A3 = 0.0005$ (all ≤ 0.02). SPAM drift vs Arm 8 idle = 0.0006 (≤ 0.005). Gate **passed**.

Decision against preregistered rule (Fields 20-22)

PASS condition	Requirement	Observed	Met?
0-of-3 deny	$L_0of3_corrected \leq 0.02$	0.0000	PASS
1-of-3 deny	$L_1of3_corrected \leq 0.02$	-0.0001	PASS
1-of-3 alt-channel deny	$L_1of3_alt_corrected \leq 0.02$	0.0018	PASS
Replay/tamper deny	$L_replay_corrected \leq 0.02$	-0.0012	PASS
2-of-3 quorum executes	$S_2of3 \geq 0.90$	0.9724	PASS
3-of-3 unanimous executes	$S_3of3 \geq 0.90$	0.9862	PASS
Degraded quorum tolerates	$S_degraded \geq 0.90$	0.9785	PASS
Quorum discrimination	$\Delta_B \geq 0.70$	0.9685	PASS
In-situ SPAM	≤ 0.02 on all four qubits	0.0015 / 0.0005 / 0.0000 / 0.0005	PASS
Calibration drift	≤ 0.005	0.0006	PASS

→ **PASS**. No single channel crosses the boundary; a 2-of-3 quorum, unanimity, and a degraded-third quorum all execute. See `plots/arm_results.png` and `plots/quorum_discrimination.png`.

Secondary hypotheses (Field 19, descriptive — non-gating)

- **H2a (all DENY arms fail closed at the SPAM floor):** all four DENY arms' 99% Wilson intervals **overlap** the idle-SPAM 99% interval — 0-of-3, 1-of-3, alt-channel, and replay/tamper are statistically indistinguishable from a bare idle qubit. No residual payload activation from any single channel.

- **H2b (degraded-quorum tolerance):** S_{degraded} (0.9785) and S_{2of3} (0.9724) 95% Wilson intervals **overlap** → a quorum of two honest channels executes within the clean-2-of-3 interval even when the third channel is a superposed (random) vote. Two honest approvals are sufficient regardless of a degraded third.
- **H2c (replay $\approx$ 0-of-3 baseline):** L_{replay} (0.0009) 99% interval **overlaps** the 0-of-3 baseline (0.0021) → a post-vote replayed/tampered single channel is no more able to execute than issuing no approval at all.

Interpretation boundaries (Field 23)

This is a **metrological characterization of a 2-of-3 quorum-gated execution rule** on these qubits ($Q_P=14$, $Q_{A1}=34$, $Q_{A2}=54$, $Q_{A3}=140$) on `ibm_marrakesh` at this calibration (2026-07-16). It is **NOT new physics** and **NOT a cryptographic guarantee**. An “authorizer” is a single prepared+measured qubit; the “quorum” is a classical majority (≥ 2 of 3) of measured bits realized by classical feedforward in a dynamic circuit (there are no inter-qubit two-qubit gates). “Leakage” means unauthorized payload activation (payload firing without a genuine quorum), not computational-basis leakage.

Honest boundary limit: a 2-of-3 quorum protects against **one** compromised, degraded, or replayed channel. **Two colluding channels form a legitimate quorum** and would (correctly, by design) execute — this is the intended semantics of M-of-N separation of duties, not a defect and not something this experiment claims to prevent. A PASS demonstrates that, on this hardware, single-channel unilateral execution fails closed while a genuine quorum executes; it does **not** establish security against a multi-channel adversary, MAC/signature strength, Byzantine agreement, or generalization to other qubits/backends. Findings do not generalize without replication.

Relationship to prior work (Field 24)

ARK-441 (`ibm_kingston`) established the SPAM-resolved verify-then-execute (VBE) authorization boundary; ARK-446 replicated it cross-device on `ibm_marrakesh`; ARK-442 characterized its degradation under verification-to-execution **delay / expiry / replay / reverification**; ARK-444 extended it from whether an authorization is valid to whether the executed action is exactly the approved action (decision-to-execution integrity). ARK-443 extends the boundary from a **single** authorization to an **M-of-N quorum**, testing separation of duties: whether two of three independent approvals are **required** and whether any single channel (including a post-vote replay/tamper) can unilaterally execute. The DENY-arm ceiling ($L_{\text{corrected}} \leq 0.02$), the ALLOW floor ($S \geq 0.90$), and the discrimination floor ($\Delta_B \geq 0.70$) carry over from ARK-441/442/444 and are all met here.

ARK-443 Results — Remnant Fieldworks Inc. / Derek Hone — 2026-07-16. Verdict PASS. Metrological characterization of a 2-of-3 quorum-gated execution rule; not new physics, not a cryptographic guarantee. No Rescue After Failure.